

MATHS SUPER FORMULAS

Dekhte Hi Answer – Verified Edition

32 Chapters • Audited Against Book + 946 JSON MCQs

946 MCQs Computer-Verified • 7 Naye Machines (Set, Log, Coded, DS,
Time-Series...)

Har formula ke saath: PEHCHAN + TERMS + SOLVED examples

Kaise Use Karein (3 Steps):

1. Question padho → **PEHCHAN** box se keywords match karo.
2. **TERMS** box dekho – har letter ka matlab wahan likha hai.
3. **SOLVED** box mein waisa hi question hal kiya hua hai – values badal kar copy karo.

Guarantee: Har formula ko book ke Types aur JSON ke 4156 MCQs ke against computer se verify kiya gaya hai. Answers match hote hain.

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Chapter 1

Number System (Sankhya Paddhati)

NS-1 — FACTOR MACHINE

PEHCHAN – Question mein ye dikhe

“number of factors / divisors”, “odd factors”, “even factors”, “kitne bhajak”, “perfect square/cube banane ke liye kya multiply/divide karein”

Number ko prime powers mein todo: $N = 2^a \times p^b \times q^c$. Phir:

$$\text{Total factors} = (a + 1)(b + 1)(c + 1)$$

$$\text{Odd factors} = (b + 1)(c + 1)$$

$$\text{Even factors} = \text{Total} - \text{Odd}$$

TERMS – Har letter ka matlab

N = diya gaya number • 2^a = number mein 2 kitni baar hai (uski power a)
• p, q = baaki odd prime numbers (jaise 3, 5, 7) • b, c = unki powers • Odd factors nikaalte waqt 2 ki power ko IGNORE kar do

SOLVED – Asli exam questions (verified)

Q (JSON): Number of divisors of 1728?

$$1728 = 2^6 \times 3^3 \Rightarrow \text{Total} = (6 + 1)(3 + 1) = \mathbf{28} \checkmark$$

Q (JSON): Factors of 400? $400 = 2^4 \times 5^2 \Rightarrow (4+1)(2+1) = \mathbf{15} \checkmark$

Q (JSON): Odd factors of $2^5 \times 3^3 \times 5^2$? 2 ignore: $(3+1)(2+1) = \mathbf{12} \checkmark$

Q (JSON): 81 ko kis number se divide karein ki perfect cube bane?

$81 = 3^4$; cube ke liye power 3 ka multiple chahiye $\Rightarrow$ ek 3 hatao $\Rightarrow$ divide by $\mathbf{3} \checkmark$

NS-2 — UNIT DIGIT MACHINE

PEHCHAN – Question mein ye dikhe

“unit digit”, “last digit”, “ikai ka ank” of a^n (koi bhi power)

Step 1: $n \div 4$ ka remainder $= r$ **Step 2:** unit digit of a^r

(Agar $r = 0$ aaye to a^4 ka unit digit lo.)

Base ka last digit	$r = 1$	$r = 2$	$r = 3$	$r = 0$
2	2	4	8	6
3	3	9	7	1
7	7	9	3	1
8	8	4	2	6
4	4	6	4	6
9	9	1	9	1
0, 1, 5, 6	Hamesha wahi digit rehta hai			

TERMS – Har letter ka matlab

a = base (jiski power hai) • n = power (exponent) • r = n ko 4 se divide karne par bacha remainder

SOLVED – Asli exam questions (verified)

Q: Unit digit of 7^{45} ? $45 \div 4$, remainder $= 1 \Rightarrow 7^1$ ka unit digit $= 7 \checkmark$

Q: Unit digit of 3^{34} ? $34 \div 4$, remainder $= 2 \Rightarrow 3^2 = 9 \Rightarrow 9 \checkmark$

NS-3 — REMAINDER MACHINE

PEHCHAN – Question mein ye dikhe

“remainder”, “sheshfal”, “divided by”, “dividend-divisor-quotient” wale sab

$$N = D \times Q + R$$

TERMS – Har letter ka matlab

N = Dividend (jo number divide ho raha hai) • D = Divisor (jisse divide kar rahe) • Q = Quotient (bhagfal) • R = Remainder (sheshfal), hamesha $0 \leq R < D$

SOLVED – Asli exam questions (verified)

Q (JSON): 156 ko 7 se divide karne par remainder?

$$156 = 7 \times 22 + 2 \Rightarrow R = 2 \checkmark$$

Q: Divisor, remainder ka 5 guna; divisor, quotient ka 2 guna; $R = 10$?

$$D = 5 \times 10 = 50; Q = \frac{50}{2} = 25 \Rightarrow N = 50 \times 25 + 10 = 1260 \checkmark$$

NS-3B — CHAIN REMAINDER MACHINE**PEHCHAN – Question mein ye dikhe**

Do divisors diye ho jinme se ek doosre ko poora divide karta ho (jaise 75 aur 15, ya 783 aur 29 ka multiple)

Agar D_2, D_1 ko poora divide karta hai ($D_1 = k \times D_2$):

$$\text{Naya Remainder} = (\text{Purana Remainder}) \bmod D_2$$

Seedha: purane remainder ko chhote divisor se divide karo, jo bacha wahi naya remainder hai.

TERMS – Har letter ka matlab

D_1 = bada divisor (jisse pehle divide kiya) • D_2 = chhota divisor (jisse ab divide karna hai, aur D_2 jo D_1 ka factor ho) • Ye tabhi chalega jab D_2, D_1 ko poora-poori divide kare

SOLVED – Asli exam questions (verified)

Q (JSON): N ko 783 se divide karne par remainder 48; 29 se divide karne par?

$$783 = 27 \times 29 \text{ (29 poora divide karta hai)} \Rightarrow 48 \bmod 29 = 19 \checkmark$$

Q (JSON): N ko 75 se divide karne par remainder 32; 15 se divide karne par?

$$75 = 5 \times 15 \Rightarrow 32 \bmod 15 = 2 \checkmark$$

NS-4 — DIVISIBILITY + POWER-IN-FACTORIAL

PEHCHAN – Question mein ye dikhe

“divisible by”, “vibhajya”, “highest power of p in $n!$ ”

Se divide	Rule
2 / 5 / 10	Last digit: even / 0 ya 5 / 0
3 / 9	Digits ka sum 3 / 9 se divide ho
4 / 8	Last 2 / last 3 digits divide ho
11	Odd-jagah digits sum – Even-jagah digits sum = 0 ya 11, 22...
6 / 12 / 15	Dono rules pass: 2×3 / 3×4 / 3×5

Highest power of prime p in $n!$ (Legendre ka formula):

$$\text{Power} = \left\lfloor \frac{n}{p} \right\rfloor + \left\lfloor \frac{n}{p^2} \right\rfloor + \left\lfloor \frac{n}{p^3} \right\rfloor + \dots$$

TERMS – Har letter ka matlab

$n!$ = factorial ($n \times (n-1) \times \dots \times 1$) • p = prime number • $\lfloor \rfloor$ = sirf poora bhagfal lo, decimal chhod do

SOLVED – Asli exam questions (verified)

Q (JSON): Highest power of 2 in 50!?

$$\left\lfloor \frac{50}{2} \right\rfloor + \left\lfloor \frac{50}{4} \right\rfloor + \left\lfloor \frac{50}{8} \right\rfloor + \left\lfloor \frac{50}{16} \right\rfloor + \left\lfloor \frac{50}{32} \right\rfloor = 25 + 12 + 6 + 3 + 1 = 47 \checkmark$$

NS-5 — SUM SERIES BOX (Ratta)

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}$$

$$1^2 + 2^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

$$1^3 + 2^3 + \cdots + n^3 = \left[\frac{n(n+1)}{2} \right]^2$$

$$\text{Pehle } n \text{ odd numbers ka sum} = n^2$$

TERMS – Har letter ka matlab

n = kitne numbers tak jodna hai • Facts: sabse chhota prime = 2; 1 na prime na composite; sabse chhota natural number = 1

NS-6 — TYRE ROTATION + NO-CYCLE HANDSHAKE

PEHCHAN – Question mein ye dikhe

“tyres equally used”, “spare tyre”, “rotated for equal wear”; ya “handshakes WITHOUT any cycle/group forming”

Tyre Rotation: r tyres ek time par use hote hain (total n tyres available), total safar D :

$$\text{Har tyre ka use} = \frac{r \times D}{n}$$

No-Cycle Handshake (tree structure):

$$\text{Max handshakes without cycle} = n - 1$$

(Normal handshake = nC_2 hota hai; “no cycle” condition ho to sirf $n - 1$)

TERMS – Har letter ka matlab

n = total tyres (ya total log) • r = ek time par kitne tyres laga hote (car ke liye 4) • D = total distance • “No cycle” ka matlab: koi bhi group wapas apne pehle wale se na mile (tree/graph jaisa)

SOLVED – Asli exam questions (verified)

Q (JSON): 5 tyres (4 use + 1 spare), 40000 km safar; har tyre kitna chala?
 $\frac{4 \times 40000}{5} = 32000$ km ✓

Q (JSON): 21 log, no-cycle handshake, maximum kitne?
 $21 - 1 = 20$ ✓

Chapter 2

Decimal Fractions (Dashamlav Bhinn)

DF-1 — COMPARE MACHINE

PEHCHAN – Question mein ye dikhe

“sabse badi/chhoti fraction”, “ascending/descending order mein lagao”

$$\frac{a}{b} \text{ vs } \frac{c}{d} \Rightarrow \text{compare } a \times d \text{ vs } b \times c$$

$a \times d$ bada $\Rightarrow \frac{a}{b}$ badi. (Cross multiply karo, bas.)

TERMS – Har letter ka matlab

a, c = numerators (upar wale) • b, d = denominators (neeche wale) • Decimals compare karne ho to barabar decimal places tak 0 laga do (1.20 vs 0.21)

SOLVED – Asli exam questions (verified)

Q (JSON): $\frac{3}{2}, \frac{4}{3}, \frac{5}{6}, \frac{9}{7}$ mein sabse badi?

$\frac{3}{2} = 1.5$ sabse aage – ya pair-wise cross multiply $\Rightarrow \frac{3}{2} \checkmark$

Q (JSON): 1.2, 2.1, 0.12, 0.21 descending?

$2.10 > 1.20 > 0.21 > 0.12 \Rightarrow \text{II} > \text{I} > \text{IV} > \text{III} \checkmark$

DF-2 — RECURRING DECIMAL MACHINE

PEHCHAN – Question mein ye dikhe

“ $0.\bar{3}$ ”, “repeating/recurring decimal ko fraction banao”

$$0.\overline{abc} = \frac{abc}{999}$$

$$0.\overline{abc} = \frac{abc - a}{990}$$

Rule: Jitne digits REPEAT hote = utne 9 lagao; jitne repeat NAHI hote = utne 0 lagao (9 ke baad).

TERMS – Har letter ka matlab

$abc =$ decimal ke baad ke digits (bar wale repeat karte hain) • $0.\overline{3}$ ka matlab $0.3333\dots$

SOLVED – Asli exam questions (verified)

$$0.\overline{3} = \frac{3}{9} = \frac{1}{3} \checkmark \quad 0.\overline{45} = \frac{45}{99} = \frac{5}{11} \checkmark \quad 0.2\overline{3} = \frac{23-2}{90} = \frac{7}{30} \checkmark$$

DF-3 — DECIMAL OPERATION MACHINE**PEHCHAN – Question mein ye dikhe**

“ $0.000121 \div 0.11$ ” type, decimal multiply/divide, cube root of decimal

Divide: decimal hatao, phir normal divide karo

Root: digits ko point se pairs todo (square: 2-2, cube: 3-3)

SOLVED – Asli exam questions (verified)

Q (JSON): $0.000121 \div 0.11?$ $\frac{0.000121}{0.11} = \frac{121}{110000} = \mathbf{0.0011} \checkmark$

Q (JSON): $\sqrt[3]{0.000729}?$ $729 = 9^3$; 6 decimal digits $\div 3 = 2 \Rightarrow \mathbf{0.09} \checkmark$

Chapter 3

Indices and Surds (Ghatank – Karani)

IS-1 — POWER LAWS

$$a^m \times a^n = a^{m+n}$$

$$\frac{a^m}{a^n} = a^{m-n}$$

$$(a^m)^n = a^{m \times n}$$

$$a^{-n} = \frac{1}{a^n}$$

$$a^0 = 1$$

$$a^{m/n} = \sqrt[n]{a^m}$$

Equation solve karna: Dono side SAME base banao, phir powers barabar rakh do.

TERMS – Har letter ka matlab

a = base • m, n = powers (ghatank) • Yaad rakho: $0.008 = 5^{-3}$, $0.25 = 2^{-2}$, $0.125 = 2^{-3}$

SOLVED – Asli exam questions (verified)

Q: $2^x = 64$? $64 = 2^6 \Rightarrow x = 6$ ✓

Q: $\frac{3^5 \times 3^4}{3^7} = 3^{5+4-7} = 3^2 = 9$ ✓

IS-2 — ROOT + SURD MACHINE

PEHCHAN – Question mein ye dikhe

“square root”, “ $\sqrt{A \pm 2\sqrt{B}}$ ”, “rationalise the denominator”

Surd todna: Do numbers m, n dhundo jinse $m + n = A$ aur $m \times n = B$:

$$\sqrt{A + 2\sqrt{B}} = \sqrt{m} + \sqrt{n}$$

$$\sqrt{A - 2\sqrt{B}} = \sqrt{m} - \sqrt{n}$$

Rationalise: $\frac{1}{\sqrt{a} + \sqrt{b}} = \frac{\sqrt{a} - \sqrt{b}}{a - b}$ (conjugate se multiply)

TERMS – Har letter ka matlab

A = root ke andar ka pehla number • $2\sqrt{B}$ = root ke andar ka root wala part
• $m > n$ rakho minus wale mein • Squares yaad: $11^2=121$, $12^2=144$, $13^2=169$,
 $14^2=196$, $15^2=225$, $16^2=256$, $17^2=289$, $18^2=324$, $19^2=361$

SOLVED – Asli exam questions (verified)

Q: $\sqrt{7+2\sqrt{12}}$? $m+n=7$, $mn=12 \Rightarrow m=4, n=3 \Rightarrow \sqrt{4}+\sqrt{3} = \mathbf{2+\sqrt{3}}$
✓

Chapter 4

Simplification (Saralikaran)

SIM-1 — BODMAS + CODED OPERATORS

PEHCHAN – Question mein ye dikhe

Lamba expression; ya “+ means $\times$, $\div$ means +...” (symbols badle hue)

Bracket $\rightarrow$ Of $\rightarrow$ Division $\rightarrow$ Multiplication $\rightarrow$ Addition $\rightarrow$ Subtraction

Coded operators: PEHLE symbols ko unke naye matlab se REPLACE karo, PHIR BODMAS lagao.

Fractions: $\frac{a}{b} \pm \frac{c}{d} = \frac{ad \pm bc}{bd}$

$$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c}$$

$$a\frac{b}{c} = \frac{ac + b}{c}$$

SOLVED – Asli exam questions (verified)

Q (JSON): + means $\times$, $\div$ means +, – means $\div$, $\times$ means –. Find $56 - 8 \times 5 \div 4 + 7$

Replace: $56 \div 8 - 5 + 4 \times 7 \Rightarrow$ BODMAS: $7 - 5 + 28 = \mathbf{30} \checkmark$

Q (JSON): $\frac{2}{5} + \frac{1}{2}?$ $\frac{2(2)+5(1)}{10} = \frac{9}{10} \checkmark$

SIM-2 — IDENTITY CANCEL TABLE

PEHCHAN – Question mein ye dikhe

Bade squares/cubes fraction mein dikhein – seedha cancel karo, calculate mat karo!

Ye dikhe	Turant likho
$\frac{a^2 - b^2}{a - b}$	$a + b$
$\frac{a - b}{a^2 - b^2}$	$a - b$
$\frac{a + b}{a^3 + b^3}$	$a + b$
$\frac{a^2 - ab + b^2}{a^3 - b^3}$	$a - b$
$a^2 + ab + b^2$	$(a + b)^2$
$a^2 + b^2 + 2ab$	$(a + b)^2$
$a^2 + b^2 - 2ab$	$(a - b)^2$
$a + b + c = 0$ ho to	$a^3 + b^3 + c^3 = 3abc$
9991×10009 type	$(A - x)(A + x) = A^2 - x^2$

TERMS – Har letter ka matlab

a, b, c = koi bhi numbers ya expressions • Pattern match karo, seedha answer likho – bada calculation kabhi mat karo

Chapter 5

LCM and HCF (L.S. – M.S.)

LH-1 — POWER RULE

Prime factorize karke:

HCF = common primes ki MINIMUM power ka product

LCM = saare primes ki MAXIMUM power ka product

Fractions ke liye: $\text{HCF} = \frac{\text{HCF of uppers}}{\text{LCM of lowers}}$; $\text{LCM} = \frac{\text{LCM of uppers}}{\text{HCF of lowers}}$

TERMS – Har letter ka matlab

HCF (M.S.) = Highest Common Factor = sabse bada number jo sabko divide kare • LCM (L.S.) = Lowest Common Multiple = sabse chhota number jo sabse divide ho jaye

SOLVED – Asli exam questions (verified)

Q (JSON): HCF of 36 and 84? $36 = 2^2 3^2$, $84 = 2^2 \cdot 3 \cdot 7 \Rightarrow \text{HCF} = 2^2 \times 3 = 12$
✓

LH-2 — APPLICATION TABLE (Kaunsa lena hai – LCM ya HCF?)

PEHCHAN – Question mein ye dikhe

“bells saath bajein”, “smallest number jo remainder de”, “largest tile/rod/cylinder”, “rows mein plant karo”

Question Pattern	Answer
Chhota number, sabse divide ho, remainder R bache	$\text{LCM} + R$
Chhota number, divide hone mein R KAM pade	$\text{LCM} - R$
Bells/lights t_1, t_2, t_3 par bajein, saath kab?	LCM baad
Bada number jo sabko divide kare	HCF
Bada, sabko divide kare, same remainder	HCF diffs
Bada number, remainders r_1, r_2, r_3 de	$\text{HCF}(a-r_1, b-r_2, c-r_3)$
Largest tile / rod / measuring cylinder	HCF

Yaad rakhne ki trick: “Chhota banana / saath aana” = LCM ; “Bada banana / baantna” = HCF

SOLVED – Asli exam questions (verified)

Q (JSON): Tanks 98, 182, 266 L – largest measuring cylinder?

$\text{HCF}(98, 182, 266) = \mathbf{14}$ L ✓

Q (JSON): Least number jo +23 karne par 42, 36, 45 se divide ho?

$\text{LCM}(42, 36, 45) = 1260 \Rightarrow 1260 - 23 = \mathbf{1237}$ ✓

Q (JSON): 8, 12, 16 ki rows mein 3 bach jate hain – kitne shrubs?

$\text{LCM}(8, 12, 16) + 3 = 48 + 3 = \mathbf{51}$ ✓

Q (JSON): Rs 16/18/20/25 ke groups mein exact batna – least amount?

$\text{LCM} = \mathbf{3600}$ ✓

LH-3 — PRODUCT LINK

$$\boxed{\text{HCF} \times \text{LCM} = N_1 \times N_2} \quad (\text{sirf DO numbers ke liye})$$

TERMS – Har letter ka matlab

N_1, N_2 = dono numbers • Teen numbers par ye formula NAHI chalta • Co-prime numbers: $\text{HCF} = 1$, $\text{LCM} = \text{product}$

SOLVED – Asli exam questions (verified)

Q: HCF= 13, LCM= 455, ek number 65, doosra?

$$\frac{13 \times 455}{65} = \mathbf{91} \checkmark$$

Chapter 6

Percentage (Pratishatata)

PC-1 — MULTIPLIER MACHINE

PEHCHAN – Question mein ye dikhe

increase / decrease / salary / price / population / “ $x\%$ of y ” / exam marks

$$\text{New} = \text{Old} \times \frac{100 \pm x}{100}$$

Ulta jana ho (New se Old): divide kar do. $x\%$ of $y = \frac{x \times y}{100}$

TERMS – Har letter ka matlab

Old = pehle wali value • New = badi/ghati hui value • x = percent change •
+ badhne par, – ghatne par

SOLVED – Asli exam questions (verified)

Q: 500 mein 20% increase? $500 \times \frac{120}{100} = \mathbf{600} \checkmark$

Q (JSON): Tree har saal $\frac{1}{8}$ badhta hai, aaj 64 cm, 2 saal mein kitna badhega?
 $\frac{1}{8} = 12.5\% \Rightarrow 64 \times \frac{9}{8} \times \frac{9}{8} = 81 \Rightarrow \text{increase} = 81 - 64 = \mathbf{17} \text{ cm} \checkmark$

Q: Pass marks 33%; mile 125, fail by 40 $\Rightarrow \text{Total} = \frac{(125+40) \times 100}{33} = \mathbf{500} \checkmark$

PC-2 — NET CHANGE MACHINE (do changes)

PEHCHAN – Question mein ye dikhe

successive increase/decrease/discount; area/revenue (do factors ka change);
VOTING: 2 candidates, winner $x\% \Rightarrow \text{margin} = (2x - 100)\% \times \text{Total votes}$

$$\text{Net}\% = a + b + \frac{a \times b}{100}$$

TERMS – Har letter ka matlab

a = pehla % change • b = doosra % change • Increase par + sign, decrease/discount par – sign lagao • Answer negative aaye = overall kami/discount

SOLVED – Asli exam questions (verified)

Q (JSON): Successive discounts 20% & 15%?

$$-20 - 15 + \frac{(-20)(-15)}{100} = -35 + 3 = -\mathbf{32\%}$$
 (yaani 32% discount) ✓

Q (JSON): Price +12% phir discount 20%?

$$12 - 20 + \frac{(12)(-20)}{100} = -8 - 2.4 = -\mathbf{10.4\%}$$
 ✓

Q (JSON): 30% discount + 20% cashback = effective?

$$-30 - 20 + \frac{600}{100} = -\mathbf{44\%}$$
 ✓

PC-3 — REVERSE MACHINE

PEHCHAN – Question mein ye dikhe

“A, B se $x\%$ zyada hai to B, A se kitna % kam?”; “price badha, consumption kitna ghatayein”

$$x\% \text{ ZYADA} \Rightarrow \text{ulta } \frac{x}{100 + x} \times 100 \% \text{ KAM}$$

$$x\% \text{ KAM} \Rightarrow \text{ulta } \frac{x}{100 - x} \times 100 \% \text{ ZYADA}$$

TERMS – Har letter ka matlab

x = diya hua percent • Zyada wale mein neeche $100 + x$; kam wale mein neeche $100 - x$ • Price-consumption wala bhi YAHİ hai (kharcha same rakhna ho to)

SOLVED – Asli exam questions (verified)

Q: A, B se 25% zyada; B, A se kitna kam? $\frac{25}{125} \times 100 = \mathbf{20\%}$ ✓

Q: Price 25% badha, kharcha same – consumption? $\frac{25}{125} \times 100 = \mathbf{20\%}$ kam ✓

Chapter 7

Profit – Loss – Discount (Labh – Hani – Chhut)

PL-1 — CP–MP–SP CHAIN

PEHCHAN – Question mein ye dikhe

cost price, selling price, marked price, profit%, loss%, discount% – kuch bhi

$$SP = CP \times \frac{100 + P}{100}$$

$$SP = MP \times \frac{100 - D}{100}$$

Dono jodo:

$$MP = CP \times \frac{100 + P}{100 - D}$$

Loss ho to P ki jagah $-L$ daal do. Chain mein kahin se enter karo, multiply/divide karke bahar aao.

TERMS – Har letter ka matlab

CP = Cost Price (kharidne ka daam) • SP = Selling Price (bechne ka daam)
• MP = Marked Price (likha hua daam / MRP) • P = Profit percent • L = Loss percent • D = Discount percent

SOLVED – Asli exam questions (verified)

Q (JSON): MP = 8000, SP = 7200, Discount%?

$$D\% = \frac{8000-7200}{8000} \times 100 = \mathbf{10\%} \checkmark$$

Q: CP = 200, Profit 15% $\Rightarrow$ SP = $200 \times \frac{115}{100} = \mathbf{230} \checkmark$

Q: 20% profit chahiye, 25% discount dena hai, CP = 100 $\Rightarrow$ MP = $100 \times \frac{120}{75} = \mathbf{160} \checkmark$

Q: 10 articles ka CP = 8 articles ka SP $\Rightarrow \frac{SP}{CP} = \frac{10}{8} \Rightarrow$ Profit = $\mathbf{25\%} \checkmark$

PL-2 — SPECIAL CASE BOX

Same SP, ek $x\%$ profit ek $x\%$ loss $\Rightarrow$ hamesha **LOSS** = $\frac{x^2}{100}\%$

Fake weight: $Profit\% = \frac{\text{Sach} - \text{Jhooth}}{\text{Jhooth}} \times 100$

TERMS – Har letter ka matlab

Sach = jo weight bola (jaise 1000 g) • Jhooth = jo asli mein diya (jaise 800 g)
 • Successive transactions par PC-2 (Net Change) hi lagta hai

SOLVED – Asli exam questions (verified)

Q: 1 kg bola, 800 g diya $\Rightarrow \frac{1000-800}{800} \times 100 = \mathbf{25\%}$ profit ✓

Q: Same SP par 10% gain + 10% loss $\Rightarrow$ Loss = $\frac{100}{100} = \mathbf{1\%}$ ✓

Chapter 8

Ratio – Proportion – Partnership

RP-1 — UNIT MACHINE

PEHCHAN – Question mein ye dikhe

Ratio diya + koi EK value di (total / difference / kisi ek ka maan) – kuch bhi nikalna ho

$$1 \text{ unit} = \frac{\text{Di hui VALUE}}{\text{Uski RATIO expression}}$$

$$\text{Answer} = (\text{jo chahiye uska ratio}) \times \text{unit}$$

TERMS – Har letter ka matlab

Value = question mein diya number (total, difference, kisi ek ki value) • Ratio-expression = us value se related ratio ka hissa (total diya to ratio ka SUM, difference diya to ratio ka DIFFERENCE)

SOLVED – Asli exam questions (verified)

Q (JSON): Sides 3:4:6, perimeter 182; largest – smallest?

$$\text{unit} = \frac{182}{3+4+6} = 14 \Rightarrow (6 - 3) \times 14 = \mathbf{42} \text{ cm } \checkmark$$

Q (JSON): Milk : Water = 3:2, water 40 L, total?

$$\text{unit} = \frac{40}{2} = 20 \Rightarrow \text{total} = 5 \times 20 = \mathbf{100} \text{ L } \checkmark$$

Q: 7:3, difference 28, chhota? $\text{unit} = \frac{28}{4} = 7 \Rightarrow 3 \times 7 = \mathbf{21} \checkmark$

Fraction-ratio ho ($\frac{1}{2}:\frac{1}{3}:\frac{1}{5}$): $\text{LCM}(2,3,5)=30$ se multiply $\Rightarrow 15:10:6$, ab normal.

RP-2 — JOIN + INVERT MACHINE

PEHCHAN – Question mein ye dikhe

$A:B$ aur $B:C$ alag diye; ya $pA = qB = rC$; ya $a\%X = b\%Y$; ya proportional

$$A:B = a:b, B:C = c:d \Rightarrow A:B:C = ac : bc : bd$$

$$pA = qB = rC \Rightarrow A:B:C = \frac{1}{p} : \frac{1}{q} : \frac{1}{r}$$

$$a\%X = b\%Y \Rightarrow X:Y = b:a$$

Proportionals: Mean = $\sqrt{ab}$; Third = $\frac{b^2}{a}$; Fourth = $\frac{bc}{a}$

TERMS – Har letter ka matlab

Join: beech wale (B) ko match karo – B ki dono values multiply karke barabar karo • Invert: coefficients ka ULTA hi ratio hai • Mean proportional of a, b = beech ka; Third: $a:b :: b:x$; Fourth: $a:b :: c:x$

SOLVED – Asli exam questions (verified)

Extension – Sum of Squares: Ratio $a:b:c$ diya, unke SQUARES ka sum diya ho:

$$(ak)^2 + (bk)^2 + (ck)^2 = \text{Diya hua sum} \Rightarrow k = \sqrt{\frac{\text{Sum}}{a^2 + b^2 + c^2}}$$

Q (JSON): Ratio 2:3:6, squares ka sum = 441; largest?

$$k = \sqrt{\frac{441}{4+9+36}} = \sqrt{9} = 3 \Rightarrow \text{largest} = 6 \times 3 = \mathbf{18} \checkmark$$

RP-3 — AGE / RATIO-SHIFT MACHINE

PEHCHAN – Question mein ye dikhe

“present ratio $a:b$; t saal baad ratio $c:d$ (ya kisi ki age given)” – AGE chapter ka master

Present values = ax aur bx maano:

$$\frac{ax + t}{bx + t} = \frac{c}{d} \Rightarrow \text{cross multiply karke } x \text{ nikalo}$$

RULE 1: t saal baad HAR aadmi ki age mein $+t$

RULE 2: n logon ke SUM mein $+n \times t$ hota hai

RULE 3: Do logon ki age ka DIFFERENCE kabhi nahi badalta

TERMS – Har letter ka matlab

$a:b$ = abhi ka ratio • $c:d = t$ saal baad ka ratio • x = unit (multiplier) • t = kitne saal aage/peeche ($-t$ peeche jane par) • n = kitne log

SOLVED – Asli exam questions (verified)

Q (JSON): Ram:Ravi = 3:4; 5 saal baad Ravi = 25; Ram?

$$4x + 5 = 25 \Rightarrow x = 5 \Rightarrow \text{Ram} = 3 \times 5 = \mathbf{15} \checkmark$$

Q (JSON): Ram:Shyam = 6:5; 7 saal baad Shyam = 32; Ram?

$$5x + 7 = 32 \Rightarrow x = 5 \Rightarrow \text{Ram} = \mathbf{30} \checkmark$$

Q (JSON): Aman:David = 2:5; 10 saal baad sum = 69; difference?

$$(2x+10) + (5x+10) = 69 \Rightarrow 7x = 49 \Rightarrow x = 7 \Rightarrow \text{diff} = 3x = \mathbf{21} \checkmark$$

Q (JSON): 3 logon ka sum aaj 90; 4 saal pehle? $90 - 3 \times 4 = \mathbf{78} \checkmark$

RP-3B — FATHER-CHILD RIDDLE (Classic Pattern)**PEHCHAN – Question mein ye dikhe**

“Father says: I was as old as you are now, when you were born”; “I was your present age at your birth”

Ye ek FIXED riddle hai – iska matlab hamesha ye hota hai:

$$\mathbf{Father_{now} = 2 \times Child_{now}}$$

Kyun: Father ki age jab child paida hua = $Father_{now} - Child_{now}$. Ye barabar hai $Child_{now}$ ke. Isliye $Father_{now} = 2 \times Child_{now}$.

TERMS – Har letter ka matlab

$Father_{now}$ = father ki abhi ki age (diya hota hai) • $Child_{now}$ = seedha aadha nikal lo • Phir kisi bhi saal aage/peeche ke liye RULE 1 lagao

SOLVED – Asli exam questions (verified)

Q (JSON): Father 42 now; son 6 saal pehle kitne ka tha?

$$Son_{now} = \frac{42}{2} = 21 \Rightarrow 6 \text{ saal pehle} = 21 - 6 = \mathbf{15} \checkmark$$

Q (JSON): Father 46 now; daughter 4 saal pehle kitni thi?

$$Daughter_{now} = \frac{46}{2} = 23 \Rightarrow 23 - 4 = \mathbf{19} \checkmark$$

RP-4 — PARTNERSHIP MACHINE**PEHCHAN – Question mein ye dikhe**

do-teen log business mein paise lagayein, profit batna hai

$$\text{Profit ratio} = C_1 \times t_1 : C_2 \times t_2 : C_3 \times t_3$$

Phir RP-1 (Unit Machine) se share nikalo.

TERMS – Har letter ka matlab

C = Capital (kisne kitna paisa lagaya) • t = kitne mahine/saal ke liye lagaya
• Beech mein paisa nikala/badhaya to time-tukde karke jodo: $C_1 t_1 + C'_1 t'_1$

SOLVED – Asli exam questions (verified)

Q (JSON): Elan 2500×8 , Jeff 2000×12 , Bill 3000×6 mah.
 $20000:24000:18000 = \mathbf{10:12:9}$ – ab unit machine se profit share ✓

Chapter 9

Work – Time – Pipe (Karya – Samay – Pipe)

WT-1 — RATE MACHINE (Work + Pipe dono ka BAAP)

PEHCHAN – Question mein ye dikhe

“A can do in x days”, together, pipe fills/empties, leak, tap – SAB

$$\text{Rate} = \frac{1}{\text{akele karne ke din/ghante}}$$

$$\text{Net Rate} = (\text{FILL walon ka jod}) - (\text{EMPTY/leak ka jod})$$

$$\text{Time} = \frac{\text{Work}}{\text{Net Rate}} \quad (\text{poora kaam/tank} = 1)$$

TERMS – Har letter ka matlab

Rate = ek din/ghante mein kitna kaam • Work = kitna kaam karna hai (poora = 1, aadha = $\frac{1}{2}$) • Fill/karne wala $\rightarrow$ rate + mein; Empty/leak/todne wala $\rightarrow$ rate – mein

SOLVED – Asli exam questions (verified)

Q: A = 10 din, B = 15 din, saath?

$$\frac{1}{10} + \frac{1}{15} = \frac{1}{6} \Rightarrow \mathbf{6 \text{ din } \checkmark}$$

Q: Pipe A fills 10 h, B empties 15 h, dono khule?

$$\frac{1}{10} - \frac{1}{15} = \frac{1}{30} \Rightarrow \mathbf{30 \text{ h } \checkmark}$$

Q: Pump 6 h mein bharta, leak ki wajah se 7 h laga; leak akela kitne mein khali karega?

$$\frac{1}{6} - \frac{1}{7} = \frac{1}{42} \Rightarrow \mathbf{42 \text{ h } \checkmark}$$

Q: A+B = 12 din, A akela = 20; B akela?

$$\frac{1}{12} - \frac{1}{20} = \frac{1}{30} \Rightarrow \mathbf{30 \text{ din } \checkmark}$$

Bech mein koi chala jaye: Done = Rate $\times$ din; Remaining = 1 – Done; naya time = $\frac{\text{Remaining}}{\text{naya Rate}}$

WT-1B — THREE-PAIR MACHINE**PEHCHAN – Question mein ye dikhe**

“A+B ek time, B+C doosra time, C+A teesra time diya ho” – individual nikalna

$(A+B)$, $(B+C)$, $(C+A)$ ke rates diye ho:

$$A + B + C = \frac{(\text{teeno rates ka jod})}{2}$$

$$A \text{ akela} = (A+B+C) - (B+C) \quad (\text{waise hi B, C ke liye})$$

TERMS – Har letter ka matlab

Har pair ki rate = $\frac{1}{\text{diya hua din}}$ • Teeno pairs jodne se har banda 2 baar count hota hai, isliye $\div 2$

SOLVED – Asli exam questions (verified)

Q (JSON): $A+B=12d$, $B+C=15d$, $C+A=20d$; A akela?

$$\text{Rates: } \frac{1}{12} + \frac{1}{15} + \frac{1}{20} = \frac{5+4+3}{60} = \frac{12}{60} = \frac{1}{5}$$

$$(A+B+C) = \frac{1}{10}; A = \frac{1}{10} - \frac{1}{15} = \frac{1}{30} \Rightarrow A \text{ akela} = \mathbf{30} \text{ din } \checkmark$$

WT-1C — MULTI-PHASE PIPE MACHINE**PEHCHAN – Question mein ye dikhe**

“pipe X, Y, Z saath khule; X kuch der baad band ho gaya, Y bhi baad mein band ho gaya” – alag-alag phases mein alag pipes chalu

Total time T maano. Har pipe jitni der CHALU rahi uska $\text{Rate} \times \text{Time}$ jodo, total = 1:

$$\sum (\text{Rate}_i \times \text{Time}_i) = 1 \text{ (poora tank)}$$

BAARI-BAARI (alternate) khole: 1 cycle (=2 ghante) ka NET kaam jodo
 $\rightarrow$ cycles repeat karo $\rightarrow$ aakhri adhoora hissa alag se gino.

TERMS – Har letter ka matlab

Agar koi pipe T khatam hone se k pehle band hui, to uska time = $(T - k)$ hai equation mein

SOLVED – Asli exam questions (verified)

Q (JSON): X,Y,Z fill 8h,12h,24h mein; X 4h pehle aur Y 3h pehle band ho gayi (total time T se pehle); T ?

$$\frac{T-4}{8} + \frac{T-3}{12} + \frac{T}{24} = 1; 24 \text{ se multiply: } 3(T-4) + 2(T-3) + T = 24$$

$$3T-12 + 2T-6 + T = 24 \Rightarrow 6T = 42 \Rightarrow T = 7 \text{ h } \checkmark$$

WT-2 — MDH MACHINE**PEHCHAN – Question mein ye dikhe**

“ M men, D days, H hours/day” type; food lasts; printers/machines/cows

$$\frac{M_1 \times D_1 \times H_1}{W_1} = \frac{M_2 \times D_2 \times H_2}{W_2}$$

Jo cheez question mein nahi hai, use 1 maan lo. Cross-multiply karo, answer aa gaya.

TERMS – Har letter ka matlab

M = men/machines/printers (kitne log) • D = days (kitne din) • H = hours per day (roz kitne ghante) • W = work/output (kitna kaam) • Subscript 1 = pehli situation, 2 = doosri situation

SOLVED – Asli exam questions (verified)

Q (JSON): 10 printers, 10 min, 10 models; 100 printers, 100 models, time?

$$\frac{10 \times 10}{10} = \frac{100 \times D}{100} \Rightarrow D = 10 \text{ min } \checkmark$$

Q: 100 aadmi, 30 din ka khana; 10 din baad 20 chale gaye; ab kitne din?

$$\text{Bacha} = 100 \times 20 = 2000 \text{ man-days} \Rightarrow \frac{2000}{80} = 25 \text{ din } \checkmark$$

Chapter 10

Mixture – Alligation (Mishran)

MX-1 — CROSS MACHINE

PEHCHAN – Question mein ye dikhe

do cheezein (price/percent) mila kar target value banana; ratio nikalna

$$\frac{\text{Sasti ki Qty}}{\text{Mehngi ki Qty}} = \frac{\text{Mehngi} - \text{Mean}}{\text{Mean} - \text{Sasti}}$$

Ula chahiye (mean nikalna ho): $Mean = \frac{Q_1 V_1 + Q_2 V_2}{Q_1 + Q_2}$

TERMS – Har letter ka matlab

Sasti/Mehngi = dono cheezon ki values (price ya %) • Mean = target/average value jo banani hai • Q = quantity, V = value • Ye weighted average ka hi ulta roop hai

SOLVED – Asli exam questions (verified)

Q: Rs 15 aur Rs 24 wali chai se Rs 20 wali banao – ratio?

$$\frac{24-20}{20-15} = \frac{4}{5} \Rightarrow 4 : 5 \checkmark$$

Q (JSON): 10000 gallon, 5% salt; 2500 gallon paani ud gaya; new %?

$$\text{Salt same} = 500; \text{new total} = 7500 \Rightarrow \frac{500}{7500} \times 100 = 6.67\% \checkmark$$

MX-2 — REPLACEMENT MACHINE

PEHCHAN – Question mein ye dikhe

“ x litre nikal kar paani bhara, ye n baar kiya”

$$\text{Pure bacha} = V \times \left(1 - \frac{x}{V}\right)^n$$

TERMS – Har letter ka matlab

V = tank/vessel ki total capacity • x = har baar kitna nikala • n = kitni baar process kiya • Pure = original cheez (milk/wine) jo ab bachi

SOLVED – Asli exam questions (verified)

Q: 40 L milk; 4 L nikal kar paani dala, 2 baar; milk bacha?

$$40 \left(1 - \frac{4}{40}\right)^2 = 40 \times 0.81 = \mathbf{32.4 \text{ L}} \checkmark$$

Chapter 11

Simple Interest (Sadharan Byaj)

SI-1 — PRT MACHINE

PEHCHAN – Question mein ye dikhe

“simple interest”, “sadharan byaj”, “per annum”, principal-rate-time mein se 3 diye ho

$$SI = \frac{P \times R \times T}{100}$$

$$A = P + SI$$

TERMS – Har letter ka matlab

SI = Simple Interest (byaj – jo extra paisa milta hai) • P = Principal (muldhan – shuru mein lagaya paisa) • R = Rate (byaj dar, % per saal) • T = Time (kitne saal) • A = Amount (total paisa = muldhan + byaj)

SOLVED – Asli exam questions (verified)

Q (JSON): SI on Rs 5000 at 4% for 3 years?

$$\frac{5000 \times 4 \times 3}{100} = \text{Rs } 600 \checkmark$$

Q (JSON): Rs 3000 at 7% for 2 years?

$$\frac{3000 \times 7 \times 2}{100} = \text{Rs } 420 \checkmark$$

SI-1B — TIME-DIFFERENCE MACHINE

PEHCHAN – Question mein ye dikhe

“same rate par T_1 saal aur T_2 saal ke SI ka difference diya ho”; “do equal sums, alag time, interest difference”

Same P , Same R , do alag time periods T_1, T_2 ka SI difference D diya ho:

$$P = \frac{100 \times D}{R \times (T_2 - T_1)}$$

(Ye seedha PRT formula hi hai – ek saal ka EXTRA interest use karo)

TERMS – Har letter ka matlab

D = do SI ka difference • $T_2 - T_1$ = kitne saal ka gap hai • “8 aur 9 saal ka SI difference” matlab $T_2 - T_1 = 1$

SOLVED – Asli exam questions (verified)

Q (JSON): 11% rate par 8 saal aur 9 saal ke SI ka difference Rs 132; sum?

$$P = \frac{100 \times 132}{11 \times 1} = \text{Rs } 1200 \checkmark$$

Q (JSON): Equal sums, 8% rate, 2 saal aur 3 saal; difference Rs 3080; sum?

$$P = \frac{100 \times 3080}{8 \times 1} = \text{Rs } 38500 \checkmark$$

SI-2 — DOUBLE-TRIPLE MACHINE

PEHCHAN – Question mein ye dikhe

“doubles itself”, “teen guna ho jata hai”, “kitne saal mein 4 guna”

$$R \times T = 100 \times (k - 1)$$

Ek multiple se doosra:

$$T_2 = T_1 \times \frac{k_2 - 1}{k_1 - 1}$$

TERMS – Har letter ka matlab

k = kitna guna hua (double = 2, triple = 3, fourfold = 4) • R = rate % • T = saal • k_1, T_1 = pehli baat; k_2, T_2 = jo poochha hai

SOLVED – Asli exam questions (verified)

Q (JSON): SI par 10 saal mein double; rate?

$$R \times 10 = 100(2 - 1) \Rightarrow R = 10\% \checkmark$$

Q (JSON): 10 saal mein double; fourfold kab?

$$T_2 = 10 \times \frac{4-1}{2-1} = 30 \text{ saal } \checkmark$$

Q (JSON): 20 saal mein double; rate? $R = \frac{100}{20} = 5\% \checkmark$

Chapter 12

Compound Interest (Chakravridhi Byaj)

CI-1 — COMPOUND MACHINE

PEHCHAN – Question mein ye dikhe

“compound interest”, “compounded annually/half-yearly/quarterly”

$$\mathbf{A = P \times \left(1 + \frac{R}{100}\right)^n} \quad \mathbf{CI = A - P}$$

Half-yearly: $R \rightarrow \frac{R}{2}, n \rightarrow 2n$. Quarterly: $R \rightarrow \frac{R}{4}, n \rightarrow 4n$.

TERMS – Har letter ka matlab

A = Amount (total paisa akhir mein) • P = Principal (muldhan) • R = Rate % per annum • n = saal (compounding periods) • CI = Compound Interest

SOLVED – Asli exam questions (verified)

Q: Rs 8000 at 10% for 2 yr CI?

$A = 8000 \times 1.1 \times 1.1 = 9680 \Rightarrow \text{CI} = 1680 \checkmark$

CI-2 — POWER MULTIPLE MACHINE

PEHCHAN – Question mein ye dikhe

“CI par double in T saal; 4 guna / 8 guna kab hoga?”

$$\mathbf{k_2 = (k_1)^x \Rightarrow T_2 = x \times T_1}$$

TERMS – Har letter ka matlab

k_1 = pehla multiple (double = 2), T_1 = uske saal • k_2 = poochha gaya multiple • x = power jisse k_1 se k_2 banta hai ($8 = 2^3$ to $x = 3$)

SOLVED – Asli exam questions (verified)**Q (JSON):** CI par 4 saal mein double; 8 times kab?

$$8 = 2^3 \Rightarrow T = 3 \times 4 = \mathbf{12} \text{ saal } \checkmark$$

Q: 3 saal mein triple; 9 times? $9 = 3^2 \Rightarrow T = 2 \times 3 = \mathbf{6} \text{ saal } \checkmark$ **CI-3 — CI-SI BRIDGE (2 saal)**

$$\text{CI} - \text{SI} = P \times \left(\frac{R}{100} \right)^2 \quad (2 \text{ saal ke liye})$$

TERMS – Har letter ka matlab

P = principal • R = rate % • Sirf 2 saal ke difference ka shortcut hai ye • 3 saal ho to: $P\left(\frac{R}{100}\right)^2\left(3 + \frac{R}{100}\right)$

SOLVED – Asli exam questions (verified)**Q:** $P = 10000$, $R = 10\%$, 2 yr; CI-SI?

$$10000 \times (0.1)^2 = \mathbf{100} \checkmark$$

CI-4 — EFFECTIVE ANNUAL RATE**PEHCHAN – Question mein ye dikhe**

“compounded half-yearly/quarterly, effective annual rate kya hai”

$$\text{Effective Rate} = \left[\left(1 + \frac{R}{100m} \right)^m - 1 \right] \times 100$$

TERMS – Har letter ka matlab

R = di gayi annual rate • m = saal mein kitni baar compound hota (half-yearly = 2, quarterly = 4) • Effective rate hamesha di hui rate se thodi zyada hoti hai

SOLVED – Asli exam questions (verified)**Q (JSON):** 10% annual, half-yearly compound; effective rate?

$$[(1.05)^2 - 1] \times 100 = \mathbf{10.25\%} \checkmark$$

Chapter 13

Average (Ausat)

AVG-1 — SUM MACHINE

PEHCHAN – Question mein ye dikhe

average ka koi bhi sawaal – SUM se khelo

$$\text{Sum} = n \times \text{Average}$$

Equally-spaced numbers (AP): Average = beech wala = $\frac{\text{pehla} + \text{aakhri}}{2}$

TERMS – Har letter ka matlab

n = kitni values • Sum = sab values ka jod • Naya banda aane par / jaane par: naya Sum nikalo, naye n se divide karo

SOLVED – Asli exam questions (verified)

Q (JSON): 3 equally-spaced numbers, average 135 $\Rightarrow$ beech wala = **135** ✓

Q: 10 innings avg 40; 11th mein kya banaye ki avg 42?

$$11 \times 42 - 10 \times 40 = 462 - 400 = \mathbf{62} \text{ ✓}$$

AVG-2 — SHIFT MACHINE

PEHCHAN – Question mein ye dikhe

“har number mein k joda/ghataya/multiply” – naya average?

$$\text{Har number} + k \Rightarrow \text{Average bhi} + k$$

$$\text{Har number} \times k \Rightarrow \text{Average bhi} \times k$$

TERMS – Har letter ka matlab

k = jo number sab mein joda/ghataya/multiply kiya • Kuch bhi calculate mat karo – seedha average par wahi operation kar do!

SOLVED – Asli exam questions (verified)

Q (JSON): 70 numbers, avg 32, har ek mein +3 $\Rightarrow$ **35** ✓

Q (JSON): 10 numbers, avg 15, har ek mein +2 $\Rightarrow$ **17** ✓

Q (JSON): 16 numbers, avg 8, har ek mein +2 $\Rightarrow$ **10** ✓

AVG-3 — REPLACE + COMBINE MACHINE**PEHCHAN – Question mein ye dikhe**

“ek aadmi replace hua, average Δ badla”; “do groups ka combined average”

$$\text{Naya aadmi} = \text{Purana aadmi} \pm n \times \Delta$$

$$\text{Combined Avg} = \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2}$$

TERMS – Har letter ka matlab

n = group mein kitne log • Δ = average kitna badla • Average badha to +, ghata to – • $\bar{x}_1, \bar{x}_2$ = dono groups ke averages; n_1, n_2 = dono ki ginti

SOLVED – Asli exam questions (verified)

Q: 8 logon mein ek (60 kg) ki jagah naya aaya, avg 2.5 badha; naya kitne ka?
 $60 + 8 \times 2.5 = \mathbf{80}$ kg ✓

Q (JSON): Class A: 49 students avg 40%; Class B: 53 avg 35%; combined?
 $\frac{49(40)+53(35)}{102} = \frac{3815}{102} \approx \mathbf{37.4\%}$ ✓

Chapter 14

Speed – Time – Distance (Chal – Samay – Duri)

STD-1 — MOTHER MACHINE

$$\text{Distance} = \text{Speed} \times \text{Time}$$

$$\text{km/h ko m/s: } \times \frac{5}{18}$$

$$\text{m/s ko km/h: } \times \frac{18}{5}$$

Average Speed (same distance dono taraf):

$$\frac{2ab}{a+b}$$

(teen barabar

legs): $\frac{3abc}{ab+bc+ca}$

TERMS – Har letter ka matlab

a, b, c = alag-alag speeds • Average speed KABHI simple average nahi hoti (jab tak time barabar na ho) • Same distance par: speed ka ratio ulta = time ka ratio

SOLVED – Asli exam questions (verified)

Q (JSON): 300 km, 5 h $\Rightarrow$ speed = **60 km/h** ✓

Q (JSON): A $\rightarrow$ B @50, B $\rightarrow$ A @60; average?

$$\frac{2 \times 50 \times 60}{110} = \mathbf{54.5 \text{ km/h}}$$
 ✓

Q (JSON): Plane, equilateral triangle, 300/200/240 km/h?

$$\frac{3 \times 300 \times 200 \times 240}{300(200) + 200(240) + 240(300)} = \mathbf{240 \text{ km/h}}$$
 ✓

STD-2 — GAP + LATE-EARLY MACHINE

PEHCHAN – Question mein ye dikhe

meet / catch / overtake; ya “speed S_1 se late, S_2 se early”

$$\text{Time} = \frac{\text{Gap}}{\text{Relative Speed}}$$

Opposite: $S_1 + S_2$ Same: $S_1 - S_2$

Late-Early distance:

$$D = \frac{S_1 \times S_2 \times (t_1 + t_2)}{S_2 - S_1}$$

TERMS – Har letter ka matlab

Gap = dono ke beech ki shuru ki doori • Relative Speed = ek doosre ke hisaab se speed (saamne se aa rahe: jodo; ek disha: ghatao) • t_1 = slow speed se kitna LATE (ghante mein); t_2 = fast speed se kitna EARLY • S_1 = slow speed, S_2 = fast speed

SOLVED – Asli exam questions (verified)

Q: 5 km/h se 7 min late; 6 km/h se 5 min early; distance?

$$D = \frac{5 \times 6 \times \frac{12}{60}}{6-5} = 6 \text{ km } \checkmark$$

Chapter 15

Train (Railgadi)

TR-1 — GOLDEN CROSSING RULE

PEHCHAN – Question mein ye dikhe

train kisi ko cross kare: pole / platform / aadmi / doosri train

$$\text{Time} = \frac{\text{Total Length to Cross}}{\text{Relative Speed}}$$

Kya cross kiya	Length	Speed
Pole / khada aadmi / signal	sirf L_T	S_T
Platform / bridge / tunnel	$L_T + L_P$	S_T
Chalta aadmi (same direction)	L_T	$S_T - S_P$
Chalta aadmi (opposite)	L_T	$S_T + S_P$
Doosri train (opposite)	$L_1 + L_2$	$S_1 + S_2$
Doosri train (same/overtake)	$L_1 + L_2$	$S_1 - S_2$

TERMS – Har letter ka matlab

L_T = train ki lambai • L_P = platform/bridge ki lambai • S_T = train ki speed
• S_P = aadmi ki speed • L_1, L_2, S_1, S_2 = dono trains ki lengths/speeds • Speed hamesha m/s mein karo ($\times \frac{5}{18}$) agar length metres mein hai

SOLVED – Asli exam questions (verified)

Q (JSON): 500 m train, pole ko 20 sec mein; speed?

$$S = \frac{500}{20} = 25 \text{ m/s} = 25 \times \frac{18}{5} = \mathbf{90 \text{ km/h}} \checkmark$$

Q (JSON): Do trains speed ratio 7:10; doosri 400 km in 5 h; pehli ki speed?

$$\text{Doosri} = 80 \Rightarrow \text{pehli} = 80 \times \frac{7}{10} = \mathbf{56 \text{ km/h}} \checkmark$$

Pole t_1 s, Platform (P m) t_2 s dono diye ho:

$$L_T = \frac{P \times t_1}{t_2 - t_1}$$

TWO PLATFORMS (pole nahi diya, seedha 2 bridges/platforms):

Bridge P_1 mein t_1 sec, Bridge P_2 mein t_2 sec:

$$\text{Speed} = \frac{P_2 - P_1}{t_2 - t_1}$$

Q (JSON): Bridge 243m/30s aur 343m/36s; speed?

$$\frac{343-243}{36-30} = \frac{100}{6} = 16.67 \text{ m/s} = \mathbf{60} \text{ km/h } \checkmark$$

Q (JSON): Bridge 1000m/120s aur 600m/80s; train length?

$$S = \frac{1000-600}{120-80} = 10 \text{ m/s}; L_T + 600 = 10 \times 80 = 800 \Rightarrow L_T = \mathbf{200} \text{ m } \checkmark$$

Chapter 16

Boat and Stream (Nav – Dhara)

BT-1 — PLUS-MINUS MACHINE

PEHCHAN – Question mein ye dikhe

upstream / downstream / still water / stream / current / dhara

$$\text{Down} = B + C$$

$$\text{Up} = B - C$$

$$B = \frac{\text{Down} + \text{Up}}{2}$$

$$C = \frac{\text{Down} - \text{Up}}{2}$$

Time ratio Up:Down = $m:n$ diya ho:

$$C = B \times \frac{m - n}{m + n}$$

TERMS – Har letter ka matlab

B = Boat ki speed still water mein (shant paani) • C = Current/stream ki speed (dhara) • Down = downstream speed (dhara ke SAATH – fast) • Up = upstream speed (dhara ke KHILAF – slow) • $m:n$ = up-time : down-time ka ratio

SOLVED – Asli exam questions (verified)

Q (JSON): Up = 14, Down = 40 km/h; stream?

$$C = \frac{40-14}{2} = 13 \text{ km/h; Boat} = \frac{54}{2} = 27 \checkmark$$

Q (JSON): Still water 4.5 km/h; up mein down se DOGUNA time; stream?

$$m:n = 2:1 \Rightarrow C = 4.5 \times \frac{2-1}{2+1} = 1.5 \text{ km/h } \checkmark$$

BT-2 — ROUND-TRIP TIME MACHINE

PEHCHAN – Question mein ye dikhe

“ D km jaake wapas aane mein total time diya ho” – same distance dono taraf, boat/still-water speed maloom, stream nikalna (ya ulta)

Same distance D dono taraf, boat speed B , stream C , total time T :

$$T = \frac{D}{B + C} + \frac{D}{B - C} = \frac{2BD}{B^2 - C^2}$$

TERMS – Har letter ka matlab

D = ek taraf ki doori • B = still water speed • C = stream speed • T = jaane + aane ka TOTAL time

SOLVED – Asli exam questions (verified)

Q (JSON): Boat still water = 20 km/h; 36 km jaake wapas = 3h 45min; stream speed?

$$3.75 = \frac{2(20)(36)}{400 - C^2} \Rightarrow 3.75(400 - C^2) = 1440 \Rightarrow C^2 = 16 \Rightarrow C = 4 \text{ km/h } \checkmark$$

Chapter 17

Permutation and Combination (Kramchay – Sanchay)

Ek hi TABLE. Question ke words dekho → row match karo → value daalo → answer.

PNC-TABLE — DEKHO – DAALO – ANSWER (12 rows)

#	Question mein ye dikhe	Formula (seedha daalo)	Example (JSON)
1	word ke letters arrange, SAB alag	$n!$	MIRACLE: $7! = 5040$
2	word mein letters REPEAT hote hain	$\frac{n!}{\text{rep}_1! \times \text{rep}_2!}$	BANANA: $\frac{6!}{3!2!} = 60$
3	kuch letters/log SAATH rakhne hain	$(\text{bachhe} + 1)! \times \text{saath!}$	LEADING vowels: $5! \times 3! = 720$
4	2 letters ENDS (extreme) pe	$2 \times (n - 2)!$	‘256974’, 6&5 ends: $2 \times 4! = 48$
5	ROUND / circular table	$(n - 1)! \quad [\text{haar ho to } \div 2]$	4 log: $3! = 6$; 6 log: $5! = 120$
6	n mein se r CHUNO (team/committee)	${}^nC_r = \frac{n!}{r!(n - r)!}$	12 Q me se 3: ${}^{12}C_3 = 220$
7	DO groups se chuno (a from m , b from n)	${}^mC_a \times {}^nC_b$	4B+2G from 8,4: $70 \times 6 = 420$
8	“at most / not more than k ”	cases $JODO$: $k=0 + k=1 + \dots$	5M11W max 3M: $= 2256$
9	X hamesha Y se PEHLE aaye	$\frac{n!}{2}$	M before N (5 log): $\frac{5!}{2} = 60$
10	digits se number, 0 se START nahi	$(n-1) \times (n-1)!$	7 digits with 0: $6 \times 6! = 4320$
11	n IDENTICAL cheezein k mein baanto	≥ 1 : ${}^{n-1}C_{k-1}$; ≥ 0 : ${}^{n+k-1}C_{k-1}$	20 toys 10 kids: ${}^{19}C_9 \checkmark$
12	handshake / sab se ek match / product-pairs	$\frac{n(n-1)}{2}$; pairs divisors $= \frac{2}{2}$	10 teams: 45 ; 840: $\frac{32}{2} = 16$

TERMS – Har letter ka matlab

n = total • r = kitne uthane • CHUNO = order matter nahi • ARRANGE = order matter hai (arrange karna ho to chunne ke baad $\times r!$) • Do kaam “AUR” se jude = MULTIPLY; “YA” se jude = ADD

SOLVED – Asli exam questions (verified)

Row 3+4 combo (JSON): 12 books, 4 hamesha saath? bundle: $9! \times 4! \checkmark$

Row 3 (JSON): 8 Indian 4 Amer 4 Eng apne-apne saath? $3! \times 8! \times 4! \times 4! \checkmark$

Row 8 (JSON): 8 ki team, 9B 7G, max 3 girls? $\sum_{g=0}^3 {}^7C_g {}^9C_{8-g} = 6435 \checkmark$

Chapter 18

Probability (Prayikta)

Ek TABLE (10 rows). Ginti karni ho to upar wala PNC-TABLE use karo.

PROB-TABLE — DEKHO — DAALO — ANSWER (10 rows)

#	Question mein ye dikhe	Formula (seedha daalo)	Example (JSON)
1	seedha probability pucha	$P = \frac{\text{fav}}{\text{total}}$	card 1-100 square: $\frac{10}{100} = \frac{1}{10}$
2	DONO/sabhi ho (“and”, independent)	MULTIPLY: $P_1 \times P_2$	dono div-by-3: $\frac{1}{3} \times \frac{1}{3} = \frac{1}{9}$
3	bina WAPAS rakhe (no replacement)	multiply, har step -1	5W3B dono W: $\frac{5}{8} \times \frac{4}{7} = \frac{5}{14}$
4	“exactly k baar in n tries”	${}^nC_k p^k q^{n-k} \quad (q=1-p)$	5 toss 3H: ${}^5C_3 (\frac{1}{2})^5 = \mathbf{0.3125}$
5	“at least one”	$1 - P(\text{ek bhi nahi})$	at least 1 six (2 dice): $1 - \frac{25}{36}$
6	“GIVEN that” / pata chal chuka	given cases mein hi $\frac{\text{fav}}{\text{total}}$	$\geq 1H$ given, dono H: $\{HH, HT, TH\} \Rightarrow \frac{1}{3}$
7	2 types/sources mile hue	$P_1 \cdot w_1 + P_2 \cdot w_2$	bulbs 50-50: $.5(.7) + .5(.4) = \mathbf{0.55}$
8	fair coin: odd/even heads; pichle results	$\frac{1}{2}$ hamesha (yaaddasht nahi)	20 toss odd H: $\frac{1}{2}$; 5th toss: $\frac{1}{2}$
9	m barabar cheezon mein se ek PEHLI ho	$\frac{1}{m}$	2 first of 10 evens: $\frac{1}{10}$
10	rate/minute + Poisson / e dikhe options mein	$\frac{e^{-\lambda} \lambda^k}{k!}$; Mean=Var= λ	mean 3, $P(X=3) = \frac{9e^{-3}}{2}$

TERMS – Har letter ka matlab

fav = jo chahiye • total = sab possible • Cards: 52; 26 red/black; 12 face • 2 dice total = 36; 3 dice = 216 • Binomial: Mean= np , Var= npq ; 1 trial: Var= pq • Given = duniya chhoti ho gayi, usi mein gino

SOLVED – Asli exam questions (verified)

Row 6 (JSON): X:5(1def)+Y:4(2def), defective mila, Y ka? def total 3, Y ke $2 \Rightarrow \frac{2}{3} \checkmark$

Row 6 (JSON): 7 bachche, youngest boy given, sab boys? baaki 6: $\frac{1}{2^6} = \frac{1}{64} \checkmark$

Row 2 (JSON): overheat 0.15, backup fail 0.11? $0.15 \times 0.11 = \mathbf{0.0165} \checkmark$

Row 4 (JSON): 3 bachche 2G+1B? ${}^3C_2(\frac{1}{2})^3 = \frac{3}{8} \checkmark$; win 0.7 Var? $0.7 \times 0.3 = \mathbf{0.21} \checkmark$

Row 10 (JSON): mean 3, $P(X < 3)$? $e^{-3}(1+3+\frac{9}{2}) = \frac{17}{2e^3} \checkmark$

PROB-THEORY — RATTA FACTS (definition MCQ)

DISCRETE	Binomial, Poisson , Bernoulli, Geometric
CONTINUOUS	Normal (std: mean 0, var 1), Exponential, Rayleigh
EXIST NAHI	“Linear distribution”
$E[X]$	$\sum x p(x)$ – har value $\times$ prob, jodo
CDF $F(x)$	left se jodte jao; aakhri pe 1
Entropy base	2=bits; e =nats; 10= bann
2 RVs ka SUM	convolution $\int_0^t f_1(x)f_2(t-x)dx$
Union bound	$P(A \cup B) \leq P(A) + P(B)$

SOLVED – Asli exam questions (verified)

Q (JSON): $E[a]$, $a=0.4$, $P=\frac{1..5}{15}$? $\frac{40}{15} = \mathbf{2.67} \checkmark$ CDF ± 1 (0.5 each)? **0.5, 1** $\checkmark$ $b=10$? **bann** $\checkmark$

Chapter 19

Mensuration (Kshetramiti)

MEN-1 — 2D TABLE

Shape	Area	Perimeter / Extra
Rectangle	$l \times b$	$2(l + b)$; diagonal = $\sqrt{l^2 + b^2}$
Square	$a^2 = \frac{d^2}{2}$	$4a$; $d = a\sqrt{2}$
Triangle	$\frac{1}{2} \times b \times h$	Heron: $\sqrt{s(s-a)(s-b)(s-c)}$
Equilateral $\triangle$	$\frac{\sqrt{3}}{4}a^2$	height = $\frac{\sqrt{3}}{2}a$
Circle	πr^2	$C = 2\pi r$
Sector (angle θ)	$\frac{\theta}{360} \times \pi r^2$	arc = $\frac{\theta}{360} \times 2\pi r$
Parallelogram	$b \times h$	$2(a + b)$
Rhombus	$\frac{1}{2}d_1d_2$	side = $\frac{1}{2}\sqrt{d_1^2 + d_2^2}$
Trapezium	$\frac{1}{2}(a + b) \times h$	a, b = parallel sides

TERMS – Har letter ka matlab

l = length, b = breadth/base, h = height, a = side, d = diagonal, r = radius, $s = \frac{a+b+c}{2}$ (semi-perimeter), d_1, d_2 = rhombus ke diagonals, $\pi = \frac{22}{7}$

SOLVED – Asli exam questions (verified)

Rectangle Shortcut – Perimeter & Diagonal diye ho, Area seedha:

$$\text{Area} = \frac{P^2}{8} - \frac{d^2}{2}$$

Q (JSON): $P = 18$, $d = \sqrt{41}$; Area?

$$\frac{18^2}{8} - \frac{41}{2} = 40.5 - 20.5 = \mathbf{20 \text{ cm}^2} \checkmark$$

MEN-2 — 3D TABLE

Shape	Volume	CSA	TSA
Cube	a^3	$4a^2$	$6a^2$
Cuboid	$l b h$	$2h(l + b)$	$2(lb + bh + hl)$
Cylinder	$\pi r^2 h$	$2\pi r h$	$2\pi r(h + r)$
Cone	$\frac{1}{3}\pi r^2 h$	$\pi r l$	$\pi r(l + r)$
Sphere	$\frac{4}{3}\pi r^3$	$4\pi r^2$	$4\pi r^2$
Hemisphere	$\frac{2}{3}\pi r^3$	$2\pi r^2$	$3\pi r^2$

TERMS – Har letter ka matlab

CSA = Curved Surface Area (sirf ghumavdar surface) • TSA = Total Surface Area (sab kuch) • l (cone) = slant height = $\sqrt{r^2 + h^2}$ • Cube diagonal = $a\sqrt{3}$

MEN-3 — SCALING RULE**PEHCHAN – Question mein ye dikhe**

“side/radius $x\%$ badha – area/volume kitna badha?”; “melt karke recast”

Length k guna ho jaye:

Perimeter k guna **Area** k^2 guna **Volume** k^3 guna

Melt/Recast: purana Volume = naya Volume

TERMS – Har letter ka matlab

k = length ka multiplier (50% badha $\Rightarrow k = 1.5$; double $\Rightarrow k = 2$) • % increase = $(k^n - 1) \times 100$

SOLVED – Asli exam questions (verified)

Q (JSON): Cube edge +50%; volume % increase?

$(1.5)^3 = 3.375 \Rightarrow \mathbf{237.5\%}$ increase ✓

Q: Radius double, area? $2^2 = 4$ guna $\Rightarrow \mathbf{300\%}$ increase ✓

Chapter 20

Algebra (Beejganit)

ALG-1 — AP – GP MACHINE

PEHCHAN – Question mein ye dikhe

same number har baar add ho (AP); same number se multiply ho (GP)

AP: $T_n = a + (n - 1)d$

$$S_n = \frac{n}{2}(\text{pehla} + \text{aakhri})$$

GP: $T_n = ar^{n-1}$

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

TERMS – Har letter ka matlab

a = pehla term • d = common difference (har baar jitna judta) • r = common ratio (har baar jitne se multiply) • T_n = n -wa term • S_n = pehle n terms ka sum

SOLVED – Asli exam questions (verified)

Q (JSON): X ki price 40 paise/saal badhe, Y ki 15 paise; 2001 mein equal kab?

Gap har saal 25 paise se band hota – initial gap se divide karo $\Rightarrow$ AP hi hai ✓

ALG-2 — IDENTITY + CHAIN MACHINE

$$(a \pm b)^2 = a^2 + b^2 \pm 2ab$$

$$a^2 - b^2 = (a - b)(a + b)$$

$$a^3 \pm b^3 = (a \pm b)(a^2 \mp ab + b^2)$$

$$a + b + c = 0 \Rightarrow a^3 + b^3 + c^3 = 3abc$$

Chain ($t = x + \frac{1}{x}$ diya ho):

$$x^2 + \frac{1}{x^2} = t^2 - 2$$

$$x^3 + \frac{1}{x^3} = t^3 - 3t$$

($t = x - \frac{1}{x}$ ho to: $x^2 + \frac{1}{x^2} = t^2 + 2$)

TERMS – Har letter ka matlab

t = di hui value of $x + \frac{1}{x}$ • Higher powers isi chain se bante jao – kabhi x mat nikalo!

ALG-3 — QUADRATIC + TWO-DIGIT MACHINE

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha \times \beta = \frac{c}{a}$$

Two-digit number: $\text{Number} = 10x + y$; reverse = $10y + x$; difference = $9(x - y)$

TERMS – Har letter ka matlab

a, b, c = equation $ax^2 + bx + c = 0$ ke coefficients • α, β = roots • x = tens digit, y = units digit • Roots se equation: $x^2 - (\text{sum})x + (\text{product}) = 0$

SOLVED – Asli exam questions (verified)

Q (JSON): 2-digit no. = $21 \times (\text{digit diff})$; har digit +2 karo to naya = purane+7 se saat zyada... $\rightarrow$ equations: $10x + y = 21(x - y)$ se solve ✓

Q (JSON): Fraction: denominator = numerator+2; dono mein +4 to $\frac{5}{6}$?
 $\frac{x+4}{x+6} = \frac{5}{6} \Rightarrow 6x + 24 = 5x + 30 \Rightarrow x = 6 \Rightarrow \text{fraction} = \frac{6}{8}$ ✓

ALG-4 — REMAINDER THEOREM

PEHCHAN – Question mein ye dikhe

“ $P(x)$ ko $(x - a)$ ya $(x + a)$ se divide karo, remainder kya hoga”

Divisor $(x - a)$: Remainder = $P(a)$

Divisor $(x + a)$: Remainder = $P(-a)$

Bas expression mein x ki jagah a (ya $-a$) daal do – pura polynomial divide karne ki zaroorat nahi!

TERMS – Har letter ka matlab

$P(x)$ = di gayi expression • a = divisor mein jo number hai (sign ulta karke plug karo) • Factor check: $(x - a)$ factor hai $\iff P(a) = 0$

SOLVED – Asli exam questions (verified)

Q (JSON): $(x^{71} + 71)$ ko $(x + 1)$ se divide; remainder?

$x = -1$ daalo: $(-1)^{71} + 71 = -1 + 71 = \mathbf{70}$ ✓

Chapter 21

Trigonometry (Trikonmiti)

TRIG-1 — VALUE TABLE

	0°	30°	45°	60°	90°
sin	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
cos	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
tan	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	∞

Yaad karne ki trick: sin row = $\sqrt{\frac{0}{4}}, \sqrt{\frac{1}{4}}, \sqrt{\frac{2}{4}}, \sqrt{\frac{3}{4}}, \sqrt{\frac{4}{4}}$; cos = iska ULTA;
tan = $\frac{\sin}{\cos}$

TRIG-2 — IDENTITY BOX

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$1 + \tan^2 \theta = \sec^2 \theta$$

$$1 + \cot^2 \theta = \csc^2 \theta$$

$$\sin(90^\circ - \theta) = \cos \theta$$

$$\tan(90^\circ - \theta) = \cot \theta$$

$$\tan \theta \times \cot \theta = 1$$

TERMS – Har letter ka matlab

θ = angle • $\sec = \frac{1}{\cos}$, $\csc = \frac{1}{\sin}$, $\cot = \frac{1}{\tan}$ • Complementary pairs (90 wale) cancel karne ke bahut kaam aate hain

TRIG-3 — HEIGHT-DISTANCE MACHINE

PEHCHAN – Question mein ye dikhe

angle of elevation / depression, tower, shadow, ladder

$$\tan \theta = \frac{\text{Height}}{\text{Base Distance}}$$

Ladder: Height = $L \sin \theta$; Base = $L \cos \theta$ (ground se angle θ)

TERMS – Har letter ka matlab

θ = elevation/depression ka angle • Height = tower/building ki unchai • Base Distance = neeche zameen par doori • L = ladder ki lambai • Shadow: $\tan \theta = \frac{\text{object height}}{\text{shadow length}}$

SOLVED – Asli exam questions (verified)

Q: Tower se 30 m door, elevation 60° ; height?

$$H = 30 \tan 60^\circ = 30\sqrt{3} \approx \mathbf{51.96} \text{ m } \checkmark$$

Chapter 22

Coordinate Geometry (Nirdeshank Jyamiti)

CG-1 — COORDINATE TOOLKIT

$$\text{Distance} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$\text{Midpoint} = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

$$\text{Slope} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$\text{Section } (m:n) = \left(\frac{mx_2 + nx_1}{m + n}, \frac{my_2 + ny_1}{m + n} \right)$$

TERMS – Har letter ka matlab

$(x_1, y_1), (x_2, y_2)$ = do points • $m:n$ = jis ratio mein point line ko kaat raha
• Parallel lines: slopes EQUAL • Perpendicular: $m_1 \times m_2 = -1$ • Line ki equation: $y = mx + c$ (c = y-intercept)

SOLVED – Asli exam questions (verified)

Dilation from Origin (factor k): $(x, y) \rightarrow (kx, ky)$

Q (JSON): Triangle $(1, 2), (3, 4), (5, 6)$; dilation factor 3 from origin?
 $(3, 6), (9, 12), (15, 18)$ ✓

Chapter 23

Geometry (Jyamiti)

GEO-1 — ANGLE SUM MACHINE

$$\triangle = 180^\circ$$

$$\text{Quadrilateral} = 360^\circ$$

$$n\text{-gon} = (n - 2) \times 180^\circ$$

$$\text{Regular polygon: har exterior angle} = \frac{360^\circ}{n}$$

$$\text{Diagonals} = \frac{n(n - 3)}{2}$$

$$\text{Similar } \triangle: \frac{ar_1}{ar_2} = \left(\frac{side_1}{side_2} \right)^2$$

$$n\text{-gon vertices se triangles} = n - 2$$

Exterior angle of $\triangle$ = dono opposite interior ka sum. Complementary = 90° ; Supplementary = 180° .

TERMS – Har letter ka matlab

n = polygon ki sides • Parallel lines cut by transversal: F/Z-shape angles EQUAL; C/U-shape = 180°

GEO-2 — CIRCLE + TRIANGLE RULE BOX

Circle rules: Centre ka angle = $2 \times$ boundary ka angle (same arc) • Semicircle mein angle = 90° • Cyclic quadrilateral: opposite angles = 180° • Tangent $\perp$ radius • Ek point se dono tangents EQUAL

Triangle rules: Pythagoras: $H^2 = P^2 + B^2$ • Triplets yaad karo: (3, 4, 5), (5, 12, 13), (7, 24, 25), (8, 15, 17), (9, 40, 41) + inke multiples • Similar triangles: sides k guna $\Rightarrow$ areas k^2 guna • Medians centroid par 2:1 katti hain • Diagonals right angle par bisect $\Rightarrow$ RHOMBUS

TERMS – Har letter ka matlab

H = hypotenuse (sabse badi side, 90° ke saamne) • P, B = baaki do sides • Median = vertex se saamne wali side ke midpoint tak

SOLVED – Asli exam questions (verified)

Q (JSON): Diagonals right angle par bisect karein to quadrilateral? **Rhombus**
✓

Q (JSON): Medians 12 & 9 perpendicular; area? $\frac{2}{3} \times 12 \times 9 = \mathbf{72} \text{ cm}^2$ ✓

Chapter 24

Statistics (Sankhyiki)

ST-1 — MEAN – MEDIAN – MODE MACHINE

$$\text{Mean} = \frac{\text{Sum}}{n}$$

Mode = sabse zyada baar

Median = sort karke beech wali value

$$\text{Mode} = 3 \times \text{Median} - 2 \times \text{Mean}$$

Normal distribution mein: Mean = Median = Mode.

TERMS – Har letter ka matlab

n = kitni values • n even ho to Median = beech ke DO ka average • Combined mean = $\frac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1 + n_2}$ • Weighted mean = $\frac{\sum wx}{\sum w}$ (w = weightage)

SOLVED – Asli exam questions (verified)

Q (JSON): Mean, least se 10 zyada, greatest se 15 kam; median = 5; teeno numbers – equations bana kar: least = $M - 10$, greatest = $M + 15$, median diya $\Rightarrow$ Sum se solve ✓

ST-2 — SPREAD MACHINE

$$\text{Variance } (\sigma^2) = \frac{\sum x^2}{n} - (\bar{x})^2$$

$$\text{SD } (\sigma) = \sqrt{\text{Variance}}$$

Har value + $k \Rightarrow$ SD SAME rehta

Har value $\times k \Rightarrow$ SD bhi $\times k$

TERMS – Har letter ka matlab

σ = standard deviation (bikhraav) • $\bar{x}$ = mean • CV = $\frac{\sigma}{\bar{x}} \times 100$; chhota CV = zyada consistent

ST-3 — TIME SERIES MACHINE (Theory MCQs)

PEHCHAN – Question mein ye dikhe

“seasonal variation”; “trend”; “residual method”; “link relative”; “time series components”

Time series ke 4 components: Trend (T) • Seasonal (S) • Cyclic (C) • Irregular (I)

- **Residual method** $\Rightarrow$ **CYCLIC** fluctuations naapta hai (Trend+Seasonal hata kar jo bacha)
- **Link Relative method** $\Rightarrow$ **SEASONAL**; consecutive periods ka data use karta hai (pura trend nahi)
- **Moving Average** $\Rightarrow$ **TREND** smooth karta hai
- Seasonal = saal ke andar repeat; Cyclic = kai saalon ka wave; Irregular = random

SOLVED – Asli exam questions (verified)

Q (JSON): Residual method measures? $\Rightarrow$ **Cyclic fluctuations** ✓

Q (JSON): Kaunsi method comparatively kam data use karti hai? $\Rightarrow$ **Link Relative** ✓

ST-4 — SKEWNESS + ENTROPY MACHINE

PEHCHAN – Question mein ye dikhe

“skewness”; “entropy of random variable”; “symmetric distribution”

$$\text{Skewness (Karl Pearson)} = \frac{\text{Mean} - \text{Mode}}{\sigma} = \frac{3(\text{Mean} - \text{Median})}{\sigma}$$

Mean > Median $\Rightarrow$ positive skew (right tail); Mean < Median $\Rightarrow$ negative skew.
Symmetric $\Rightarrow$ skewness = 0.

$$\text{Entropy } H(X) = - \sum_{i=1}^n p(x_i) \log p(x_i)$$

$$\mathbf{b}_{yx} \times \mathbf{b}_{xy} = \mathbf{r}^2 \Rightarrow \text{dono regression coefficients ka SIGN SAME; } 0 \leq r^2 \leq 1$$

Maximum entropy tab jab sab outcomes equally likely: $H_{max} = \log n$.

TERMS – Har letter ka matlab

$p(x_i)$ = probability of value x_i • σ = SD • Entropy = uncertainty/randomness ka measure (CS/IT wale papers mein aata hai)

ST-5 — GROUPED DATA MACHINE (table wale – book gap)

PEHCHAN – Question mein ye dikhe

class-interval table dikhe (0–50, 50–100...); “median class”; “bahulak/mode”; “mean deviation”

$$\text{Mode} = L + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h$$

$$\text{Median} = L + \frac{\frac{N}{2} - CF}{f} \times h$$

$$\text{Median class: CF pehli baar} \geq \frac{N}{2}$$

$$\text{Mean Dev.} = \frac{\sum |x - \bar{x}|}{n}$$

TERMS – Har letter ka matlab

L = us class ki NEECHE wali limit • f_1 = sabse badi frequency (Mode wali class) • f_0/f_2 = uske upar/neeche wali • h = class ki chaudi • CF = median class se PEHLE tak ka jod

SOLVED – Asli exam questions (verified)

Q (Book p545): $L=40, f_1=70, f_0=38, f_2=42, h=10$? $40 + \frac{32}{60} \times 10 = 45.33 \checkmark$

Q (Book p544): attendance table? $14 + \frac{6}{22} \times 4 = 15.09 \checkmark$ **Q (Book p534):** $n=50, 25\text{th}=11, 26\text{th}=16$? $\frac{11+16}{2} = 13.5 \checkmark$

Chapter 25

Clock and Calendar (Ghadi – Panchang)

CL-1 — CLOCK ANGLE MACHINE

PEHCHAN – Question mein ye dikhe

“angle between hands at H:M”; “hands coincide/opposite kab”

$$\theta = \left| 30 \times H - \frac{11 \times M}{2} \right|$$

Agar $\theta > 180^\circ$ aaye to $360^\circ - \theta$ lo.

Coincide: $M = \frac{60H}{11}$ minutes past H .

TERMS – Har letter ka matlab

θ = dono suiyon ke beech ka angle (degrees) • H = hours (ghanta) • M = minutes • Coincide = dono suiyaan ek jagah

SOLVED – Asli exam questions (verified)

Q (JSON): 4:20? $|120 - 110| = 10^\circ$ ✓

Q (JSON): 6:30? $|180 - 165| = 15^\circ$ ✓

Q (JSON): 8:30? $|240 - 165| = 75^\circ$ ✓

Q (JSON): 9:20? $|270 - 110| = 160^\circ$ ✓

Q (JSON): 6 aur 7 ke beech coincide? $M = \frac{360}{11} = 32\frac{8}{11}$ min past 6 ✓

CL-2 — CALENDAR MACHINE

Day shift = (total days) $\div$ 7 ka remainder

Ordinary year = 1 odd day; Leap year = 2 odd days. Leap check: $\div 4$ (century ho to $\div 400$).

TERMS – Har letter ka matlab

Odd days = 7 se divide karne par bache din • 100 saal = 5 odd days; 400 saal = 0 (calendar repeat!)

Chapter 26

Set Theory and Logarithm (Samuchchay – Loghuganak)

SET-1 — POWER SET + VENN MACHINE

PEHCHAN – Question mein ye dikhe

“number of subsets”; “power set”; “ $n(A \cup B)$ ”; “kitne log dono khelte”

Subsets of n -element set = 2^n

Proper subsets = $2^n - 1$

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

$$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C) + n(A \cap B \cap C)$$

Venn diagram = logical relations ko circles se dikhana; ye PROOF nahi hota, sirf illustration hota hai.

TERMS – Har letter ka matlab

$n(A)$ = set A ke elements • $\cup$ = union (ya) • $\cap$ = intersection (aur dono) •
Power set = sabhi subsets ka set

SOLVED – Asli exam questions (verified)

Q (JSON): $A = \{x, y, z\}$, power set mein kitne subsets? $2^3 = 8 \checkmark$

LOG-1 — LOG RULES MACHINE

PEHCHAN – Question mein ye dikhe

“log” dikhe question mein; “ $\log_a x$ ”; entropy formulas

$$\log(ab) = \log a + \log b$$

$$\log \frac{a}{b} = \log a - \log b$$

$$\log a^n = n \log a$$

$$\log_a a = 1$$

$$\log 1 = 0$$

$$\log_a x = \frac{\log x}{\log a} \text{ (base change)}$$

TERMS – Har letter ka matlab

Base same hona zaroori multiply/divide rules ke liye • $\log_2$ CS questions mein (bits/entropy) • $\log_{10} 2 \approx 0.301$, $\log_{10} 3 \approx 0.477$

Chapter 27

Coded Operations and Odd One Out

COD-1 — SYMBOL SWAP MACHINE

PEHCHAN – Question mein ye dikhe

“If $+$ means $\times$, $-$ means $\div$...” – symbols ka matlab badla ho

2-Step Method:

Step 1: Har symbol ko naye matlab se REPLACE karo

Step 2: Ab normal BODMAS lagao

BODMAS order: Bracket $\rightarrow$ Of $\rightarrow$ Division $\rightarrow$ Multiplication $\rightarrow$ Addition $\rightarrow$ Subtraction.

SOLVED – Asli exam questions (verified)

Q (JSON): $+$ means $\times$, $-$ means $\div$, $\times$ means $-$, $\div$ means $+$: $16 \times 4 + 12 - 4 = ?$
Replace: $16 - 4 \times 12 \div 4 \Rightarrow$ BODMAS: $16 - (4 \times 3) = 16 - 12 = 4 \checkmark$

ODD-1 — ODD ONE OUT MACHINE

PEHCHAN – Question mein ye dikhe

“Identify the odd one”; “one number is not like the others”

Pattern check karne ka ORDER (jaldi wale pehle):

1. **Prime?** – baaki sab prime, ek composite (ya ulta)
2. **Square/Cube?** – 121, 144, 169; ya $1331=11^3$, $2197=13^3$, $3375=15^3$
3. **Even/Odd** – ek hi even ya ek hi odd
4. **Difference pattern** – consecutive gaps ($+2, +4, +6...$)
5. **Digit sum / divisibility** – sab 3 se divisible, ek nahi

SOLVED – Asli exam questions (verified)

Q (JSON): 2, 3, 5, 7, 11, 13, 17, 20 $\Rightarrow$ sab prime, **20** composite ✓

Q (JSON): 1331, 2197, 3375, 4096 $\Rightarrow$ pehle teen ODD numbers ke cube ($11^3, 13^3, 15^3$), **4096** = 16^3 even ka cube ✓

Chapter 28

Data Sufficiency (Aankda Paryaaptata)

DS-1 — STATEMENT SUFFICIENCY MACHINE

PEHCHAN – Question mein ye dikhe

“Statements I and II diye hain”; options mein “alone sufficient / both necessary”

Har statement ko AKELE test karo, phir DONO ko saath:

1. Sirf I se unique answer nikla? $\Rightarrow$ “I alone sufficient”
2. Sirf II se nikla? $\Rightarrow$ “II alone sufficient”
3. Dono akele fail, par saath mein nikla? $\Rightarrow$ “Both together needed”
4. Saath mein bhi nahi nikla? $\Rightarrow$ “Data insufficient”

GOLDEN RULE: Answer ki VALUE nahi puchi jaati – sirf ye ki answer NIKAL SAKTA hai ya nahi. Solve mat karo, sirf check karo ki equations kaafi hain ya nahi (unknowns $\leq$ equations?).

SOLVED – Asli exam questions (verified)

Q (JSON): 10 poles ka total weight? (I) one-fourth poles ka weight X kg; (II) har pole same weight.

I akela: baaki poles ka pata nahi $\times$; II akela: koi number hi nahi $\times$; DONO: $\frac{10}{4}$ poles = $X \Rightarrow 1$ pole = $\frac{4X}{10}$, total = $4X \checkmark \Rightarrow$ **Both needed**

Chapter 29

Data Interpretation (Samank Vishleshan)

DI mein ZERO naya formula! Sirf ye 3 purane lagte hain:

1. **PC-1:** $\% = \frac{\text{Part}}{\text{Whole}} \times 100$; $\% \text{ change} = \frac{\text{New} - \text{Old}}{\text{Old}} \times 100$
2. **RP-1:** Ratio of A to B = $\frac{A}{B}$ simplify karo
3. **AVG-1:** $\text{Sum} = n \times \text{Average}$

Tips: Units dhyan se (thousands/lakhs likha hota hai). Options door ho to approximate karo.

MASTER SUMMARY – Ek Nazar Mein Sab

Chapter	Kitne	Core yaad rakho
Number System	7	$(a+1)(b+1)(c+1)$; cyclicity; chain-rem; tyre
Decimal Fractions	3	ad vs bc ; 9-0 rule
Indices–Surds	2	power laws; $m+n=A$, $mn=B$
Simplification	2	BODMAS; cancel table
LCM–HCF	3	min/max power; $LCM+R$; $HCF \times LCM$
Percentage	3	$\times \frac{100 \pm x}{100}$; $a+b+\frac{ab}{100}$; $\frac{x}{100 \pm x}$
Profit–Loss–Discount	2	CP→MP→SP chain; $\frac{x^2}{100}$
Ratio–Age–Partnership	6	unit machine; sum-of-squares; father-riddle; $C \times t$
Work–Pipe	4	Rate = $\frac{1}{din}$; 3-pair; multi-phase; MDH
Mixture	2	cross; $(1 - \frac{x}{V})^n$
SI	3	$\frac{PRT}{100}$; time-diff; $RT = 100(k-1)$
CI	4	$P(1+\frac{R}{100})^n$; power rule; $P(\frac{R}{100})^2$; eff.rate
Average	3	Sum = $n\bar{x}$; shift; replace
Speed–Train–Boat	6	$D=ST$; harm-mean; platform; round-trip
P & C	1	DEKHO-DAALO table (12 rows)
Probability	2	DEKHO-DAALO table (10 rows) + facts
Mensuration	3	2D table (+P,d shortcut); 3D table; k, k^2, k^3
Algebra	4	AP/GP; identities; quadratic; remainder theorem
Trigonometry	3	value table; $\sin^2 + \cos^2 = 1$; $\tan = \frac{H}{D}$
Coordinate	1	distance-midpoint-slope kit; dilation
Geometry	2	angle sums; circle+triplet rules
Statistics	5	3Med–2Mean; SD; grouped Mode/Median; MD
Set–Log	2	2^n subsets; $n(A \cup B)$; log rules
Coded–OddOne	2	replace→BODMAS; prime/cube check
Data Sufficiency	1	I alone / II alone / both / neither
Clock–Calendar	2	$ 30H - \frac{11M}{2} $; odd days
DI	0	PC-1 + RP-1 + AVG-1
TOTAL ≈ 82		32 chapters – audited + compressed

EXAM STRATEGY

Question padho $\rightarrow$ PEHCHAN se keyword match $\rightarrow$ TERMS dekh
kar values pehchano $\rightarrow$ Formula mein daalo $\rightarrow$ Answer.

Phas jao to: Ratio banao $\rightarrow$ Unit nikalo $\rightarrow$ Multiply karo.